MTH202 QUIZ NO: 4 SOLUTION FILE

EFFORT BY: ABU HURAIRA FARAZ

Whatsapp: 0344 - 6224796

1. What is the probability of getting a number greater than 5 when a dice is tossed? **Answer: 1/6**
2. Suppose that A and B are events in a sample space. If A and b disjoint, could P (A) = 0.6 and P (B) = 0.5? **Answer: No**
3. The ____ of the experiment is the set of possible outcomes. **Answer sample space**
4. An event is a subset of? **Answer: sample space**
5. A boy have 3 red and 4 blue balls, what is the probability that a ball chosen from the boy is blue? **Answer: 3/7**
6. Among 20 people, 15 either swim or jog or both. If 5 swim and 6 swim and jog. How many jog? **Answer: 16.**
7. An urn contains six red and nine blue balls. What is the probability that a ball chosen from the urn is blue? **Answer: 9/15**
8. _____ is a procedure that yields a given set of possible outcomes? **Answer: experiment**
9. How many integers from 1 through 1002 are multiples of 4? **Answer: 250.5**
10. The probability of a an event lies between 0 and 1. **Answer: true**
11. When a dice and a coin are tossed together, then which of the following is the sample space? **Answer: {1H, 2H, 3H, 4H, 5H, 6H, 1T, 2T, 3T, 4T, 5T, 6T}**
12. If two light bulbs are chosen at random from 5 bulbs of which 3 are defective, then which of the following is the probability that none is defective? **Answer: 2/8**
13. If A and B are finite sets then the mathematical representation of inclusion-exclusion principle. **Answer: n (AUB) = n (A) + n (B) – n($A \cap B$).**
14. Let n (A) = 108, n (B) = 72, and n ($A \cup B$) = ____ **Answer: 144**
15. A tree is normally constructed from ____. **Answer: left to right**
16. Let a and B be subsets of U with n (A) = 12, n (B) = 15, n (A') = 17, and n (A intersection B) = 8, then n (U) = ____. **Answer: 29**
17. A boy have 5 res and 7 blue balls, what is probability that a ball chosen from the boy is red? **Answer: 5/12**
18. If A and B are finite set then n (AUB) = ____. **Answer: n (A) +n (B) – n($A \cap B$)**
19. If A and B are disjoint finite sets then n (AUB) = ____. **Answer: n (A) + n (B)**
20. There are 5 girls' students and 20 boys students in a class. How many students are there in total? **Answer: 25**
21. If A is only set and U is the universal then N (A') = ____. **Answer: n (U) – n (U∩A)**
22. If A and B are disjoint sets then the mathematical representation of inclusion-exclusion principle. **Answer: n (AUB) = n (A) + n (B)**
23. If A and B are disjoint sets then P (A B) =…… **Answer: P (A)**
24. If B is a subset of A then P(A/B)= 1 **Answer: True**
25. P(AC /B)=1 - P(A/B) **Answer: True**

26. If A and B be events with P (A) =1/3. P (B) =1/4 and P (A intersection b)=1/6, then P(A U B) = .
Answer: 5/12
27. P (A/B) = P (A n b)/P (B), where . **Answer: P (B) > 0**
28. If A and B be events with P (A) =1/3. P(B)=1/4 and P(A intersection b)=1/6, then P(A/B) = **Answer: 2/3**
29. Conditional probability of A given B is defined as . **Answer: P (A n B) / P (B)**
30. If X and Y are independent random variables and a and b are constant, then Var(aX+bY) is equal to
Answer: A^2 Var(X) + b^2Var(Y)
31. If the random variable X denotes the number of heads when three distinct coins are tossed, then X assumes the value **Answer: 0,1,2,3**
32. Which of the followings is the probability of getting a 6 when a dice is tossed? **Answer: 1/6**
33. Conditional probability of B given A is defined as . **Answer: P (A n B) / P (A)**
34. If A is a subset of B then P(A I B) = P(A)/P(B) **Answer: True**
35. What is the probability of the number of one head when two fair coins are tossed? **Answer: 2/4**
36. If A and B are disjoint sets then P (A/B) = ? **Answer: P (A)**
37. Which of the followings is the product set A*B*C? where A= {a}, B={b}, and C={c,d}
Answer: {(a, b, c), (a, b, d)}
38. Among 20 people, 15 either swim or jog or both. If 5 swim and 6 swim and jog, how many jog? **Answer: 16**
39. Let X= {1, 2, 3}, then 2-combinations of the 3 elements of the set X are ? **Answer: {1, 2}. {1, 3} and {2, 3}**
40. P (A n B) = P (A). P(A/B) **Answer: False**
41. If B is a subset of A then P(A/B) = 1 **Answer: True**
42. An expected value of a random variable is equal to its mean **Answer: True**
43. If A and B are two independent events then P(A n b) = **Answer: P (A) P (B)**